Ali R. Vahdati

EVOLUTIONARY BIOLOGIST · COMPUTATIONAL BIOLOGIST

Zurich, Switzerland

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Key Skills

Analytical	Biological modeling, Machine learning, Statistics, Mathematical optimization, Agent-based modeling, Working with large genomic datasets, Population genetics
Programming	Julia, Python, R, Nim, LaTeX, MySQL, SQLite, Git, Continuous integration, Unit testing, Docker, Singularity
Visualization	Vega-lite, Matplotlib, Processing, Makie.jl, Principles of effective visualization
Soft	Resilience, Leadership, Commitment, Adaptability
Transferable	Teaching, Presentation, Writing, Time management, Project management
Languages	English (fluent), Persian (native), German (beginner)

Work Experience

Consultant — *Busan, South Korea*

IBS center for climate physics, University of Pusan — *May and June 2022*

- I built a model to analyze the effect of climate on human genetic diversity

Senior lecturer — *Zurich, Switzerland*

Christoph Zollikofer group, Department of Anthropology, University of Zurich — *Aug. 2020 - Current*

- I study the ancient human dispersals and extinctions using simulations.

Postdoctoral researcher — *Zurich, Switzerland*

Christoph Zollikofer group, Department of Anthropology, University of Zurich — *Sep. 2017 - Aug. 2020*

- I developed research projects and analyzed results using statistical tools and machine learning.

International collaborations

Modeler — *Madeira, Portugal*

University of Madeira — *2022*

- I built a model for understanding metacommunity dynamics of complex life-cycles in exploited seascapes.

Modeler — *Mérida, Mexico*

Centro de investigación Cientifica de Yucatán — *Oct. 2021*

- I built statistical models for understanding interaction strengths in food webs.

Education

University of Zurich — *Zurich, Switzerland*

PhD in Evolutionary Biology — *Jan. 2012 - Sep. 2017*

- Thesis: Genotype networks: convergent evolution, population size and adaptation.
- Published four scholarly papers.

University of Manchester — *Manchester, UK*

M.S. in Evolutionary Biology and Genomics — *Sep. 2010 - Sep. 2011*

- Thesis 1: Identifying QTL and genomic imprinting effects on mice tissue phenotypes.
- Thesis 2: Analysis of replication timing on genetic variation in the human genome.
- Published one scholarly paper.

Selected Peer-reviewed Papers

EvoDynamics.jl: a framework for modeling eco-evolutionary dynamics *Journal of Open Source Software*
R Vahdati A, Melián C J *2022*

Exploring Late Pleistocene hominin dispersals, coexistence and extinction with agent-based multi-factor models *Quaternary Science Reviews*
R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE *2022*

Agents.jl: A performant and feature-full agent based modelling software of minimal code complexity *Simulation*
Datseris G, R Vahdati A, DuBois T C *2022*

Drivers of Late Pleistocene human survival and dispersal: an agent-based modeling and machine learning approach *Quaternary Science Reviews*
R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE *2019, 221*

Agents.jl: agent-based modeling framework in Julia *Journal of Open Source Software*
R Vahdati, A *2019, 4(42), 1611*

Population size affects adaptation in complex ways: simulations on empirical adaptive landscapes. *Evolutionary Biology*
Vahdati AR, Wagner A *2017*

Effect of Population Size and Mutation Rate on the Evolution of RNA Sequences on an Adaptive Landscape Determined by RNA Folding *International Journal of Biological Sciences*
Vahdati A R, Sprouffske K, Wagner A *2017;13(9):1138–51*

Parallel or convergent evolution in human population genomic data revealed by genotype networks *BMC Evolutionary Biology*
Vahdati A R, Wagner A *2016*

VCF2Networks: applying Genotype Networks to Single Nucleotide Variants data *Bioinformatics*
Dall'Olio G M, Vahdati A R, Bertranpetit J, Wagner A, Laayouni H *2014*

Genomic imprinting and genetic effects on muscle traits in mice *BMC Genomics*
Kaerst S*, Vahdati A R*, Brockmann G, Hager R *2012*
* Equal contribution

Teaching Experience

BIO397 - Applied Machine Learning - block course *University of Zurich*
Lecturer *2019-2022*
- I have designed and delivered lectures and exercises on Julia programming language, linear algebra, multivariate calculus, machine learning algorithms, neural networks, and evaluation and improvement of machine learning models.
- I have prepared students with no previous knowledge of machine learning to be able to design and perform complete machine learning workflows.

An introduction to agent-based modeling *University of Pusan, Busan, Switzerland*
Lecturer *2022*
- I gave a lecture and hands-on tutorial on using Julia language for building agent-based models.

Reproducible research - workshop *EAWAG – Kastanienbaum, Switzerland*
Lecturer *2018*
- I gave a lecture and hands-on tutorial on using Git version control software and collaborating using Github.

Hobbies and interests

Programming *github.com/kavir1698*

2015-PRESENT

- Wrote the first agent-based modeling framework in the Julia language (Agents.jl) - 2019.
- Built a GUI application for organizing and quickly accessing research notes (`GraphNotes`) - 2019.
- Created a thesaurus of Persian nouns from Moeen Dictionary (`Moeen_thesaurus`) - 2018.
- Created a program to analyze photography habits using the metadata of taken photos (`AnalyzeImageMetadata.jl`) - 2018.
- Currently developing an agent-based framework for population genetics modeling.

Linguistics and translation

- Translated into Persian and published Karl Popper's "Three worlds" - 2017.
- Created a thesaurus of Persian nouns using the Moeen dictionary - 2018.

Music

2014-Present

- I play Setar and write melodies. I enjoy the creative process of writing music.

Photography *instagram.com/alirvahdati*

2012-PRESENT

- I learn to see through photography and drawing. It is a meditative exercise for me.